Tides

Course : CE60222

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Gravity

Tides

- A tide is the **gradual oscillation** of the water surface in the ocean, which is caused by the **gravitational attraction** between the sun, moon and the earth.

- Tides produce **regular water level variations** in oceans and coastal regions.

- Tides **strongly influence** coastal hydrodynamics, sediment transport, navigation, and coastal structures.

- Essential for **harbor design, coastal modeling, and sediment transport studies.**

Generation of the tide

Equilibrium theory of the tide

- **Isaac Newton** first explained **the generation of tides.**

- He assumed that ocean water responds **instantaneously to tidal forces**, forming the basis of the **equilibrium theory of tides**.

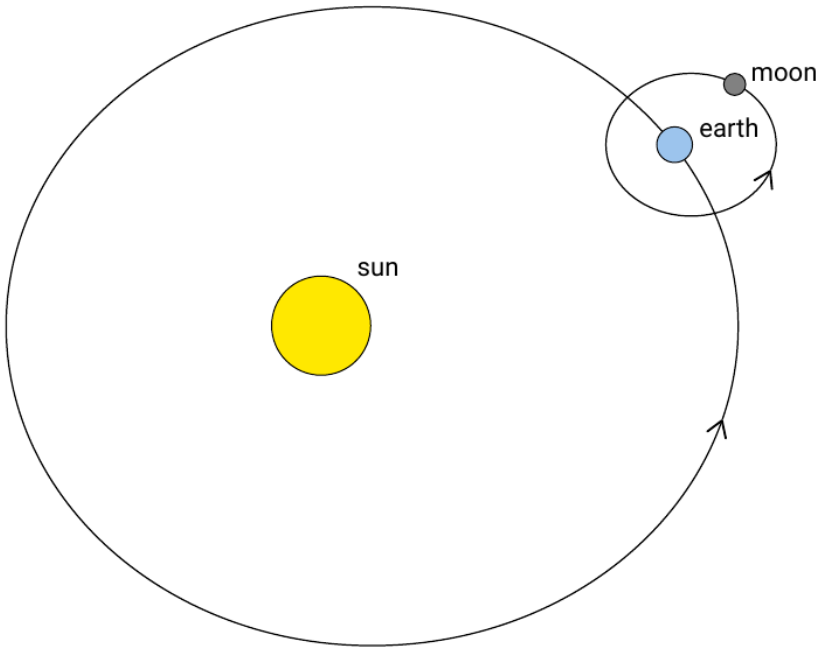
- In this theory, the Earth is considered **fully covered by water and the influence of continents is neglected**.

Gravitational pull

- Tides are generated by the **gravitational attraction** of the Moon and the Sun on Earth's oceans.

- The **Earth–Sun** and **Earth–Moon** systems rotate around a **common center of mass**, and the gravitational force between them depends on their **masses and the square of their distance**.

- The difference in mass between the sun and the moon causes the common center of mass of the earth-sun system to be located inside the sun, and of the earth-moon system within the earth.



- The **gravitational pull of the sun on 1 kg of mass of the earth** using the average distance between the sun and the earth of $d_s = 1.5 \times 10^{11}$ m is:

$$a_s = G\frac{M_s}{d_s^2} = 6.0 \times 10^{-4}\ \mathrm{g} \qquad (1)$$

where $G = 6.6 \times 10^{-11} N\, m^2 /kg$ = the universal gravitational constant and
$g = 9.81\ N/kg$ (or equivalently m/s^2)= the gravitational pull of the earth itself at the surface of the earth.

- Similarly, **the gravitational pull of the moon on 1 kg of mass of the earth** at an average distance of $d_m = 3.84 \times 10^8$ m is:

$$a_m = G\frac{M_m}{d_m^2} = 3.4 \times 10^{-6}\ \mathrm{g} \qquad (2)$$

- The **gravitational pull** of the sun/moon provides the **centripetal acceleration** that keeps Earth in orbit around the **shared center of mass**.

- This **centripetal acceleration** is the same everywhere on Earth — parallel to the Earth-Sun (or Earth-Moon) line, with magnitude equal to the **gravitational acceleration** at Earth's center.

- However, at **no point on Earth's surface** does the local gravitational acceleration **exactly match** this **centripetal acceleration** in both **magnitude and direction.**

Differential pull or the tide-generating force

- The **Sun's gravitational** pull is much **stronger** than that of the Moon, but it **contributes** only about **30% of tidal amplitudes**, whereas the **Moon accounts for about 70%.**
- This is because tides are generated **not by the total gravitational pull**, but by the **differential pull or tide-generating forces.**
- **Gravitational attraction** provides the **centripetal acceleration** that keeps the Earth moving around **the common center of mass** of the Earth–Sun and Earth–Moon systems.
- After accounting for this overall attraction, tides are generated by the **difference in gravitational pull on ocean water at different distances from the Sun and the Moon**.

- **Consider 1 $\boldsymbol{kg}$ of mass** on the near side of the earth which therefore is the earth's radius $R = 6.37 \times 10^6\ m$ closer to the sun than the center of the earth is.

- The **gravitational pull of the sun on 1 $\boldsymbol{kg}$ of mass on the near side of the earth** is Δa_s greater than a_s , i.e.:

$$\Delta a_s = a_{s,\,\text{near side}} - a_s = G\frac{M_s}{(d_s - R)^2} - G\frac{M_s}{d_s^2} \approx$$

$$2G\frac{M_s R}{d_s^3} = a_s\frac{2R}{d_s} = 0.515 \times 10^{-7}\ \text{g} \qquad (3)$$

- This is known as **differential pull**. Note that the first-order approximation of Δa_s is inversely proportional to the cube of the distance. Similarly for the moon:

$$\Delta a_m \approx 2G\frac{M_m R}{d_m^3} = a_m\frac{2R}{d_m} = 1.13 \times 10^{-7}\ \text{g} \qquad (4)$$

- The **point furthest** from the sun (moon) is in **turn smaller** than a_s (a_m) by the **same amount** Δa_s (Δa_m) which can be shown by a similar calculation.

- Carrying out the calculation for all places on the earth leads to a differential pull on the earth, as indicated schematically in Figure 3(compare it with Figure 2).

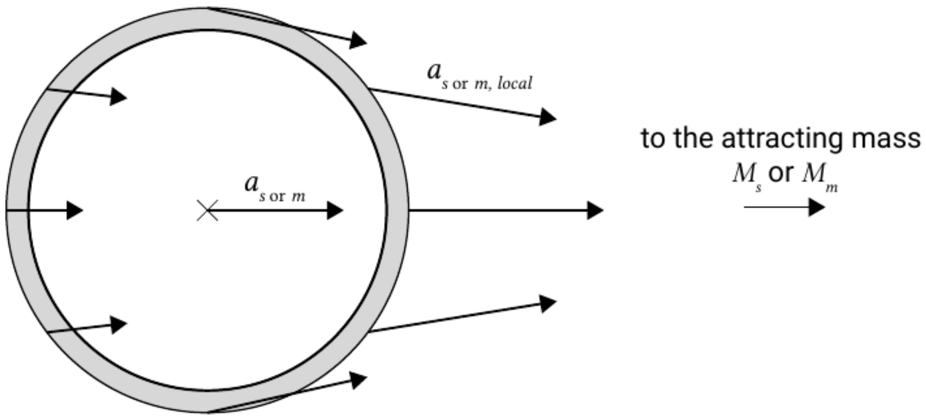


Figure 2: The gravitational acceleration towards the sun a_s, local or moon a_m, local depends on the distance to the attracting mass and on the angle to its center.

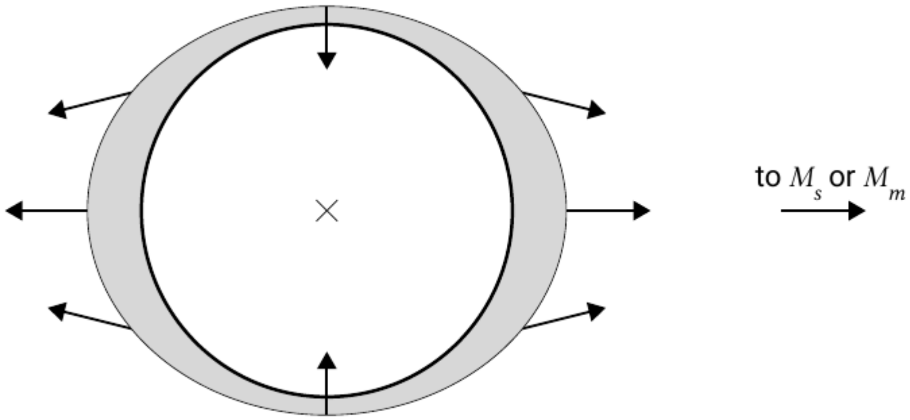


Figure 3: The direction and relative magnitudes of the differential gravitational pull at different locations on the earth's surface.

- The **differential pull** is responsible for the **tidal generation** and is therefore also referred to as **tidal force.**

- Since $\boldsymbol{\Delta a}$ **is proportional to** $\boldsymbol{M/d^3}$, the solar differential pull Δa_s is only 0.46 times the lunar differential pull Δa_m . The **moon is responsible for 69** % of the tidal mechanism, as Δa_m /(Δa_m + Δa_s) = 69 %.

- The tangential or 'horizontal' components are of the same order of magnitude as the normal components, but since they are perpendicular to the earth's gravity field, they cannot be neglected.

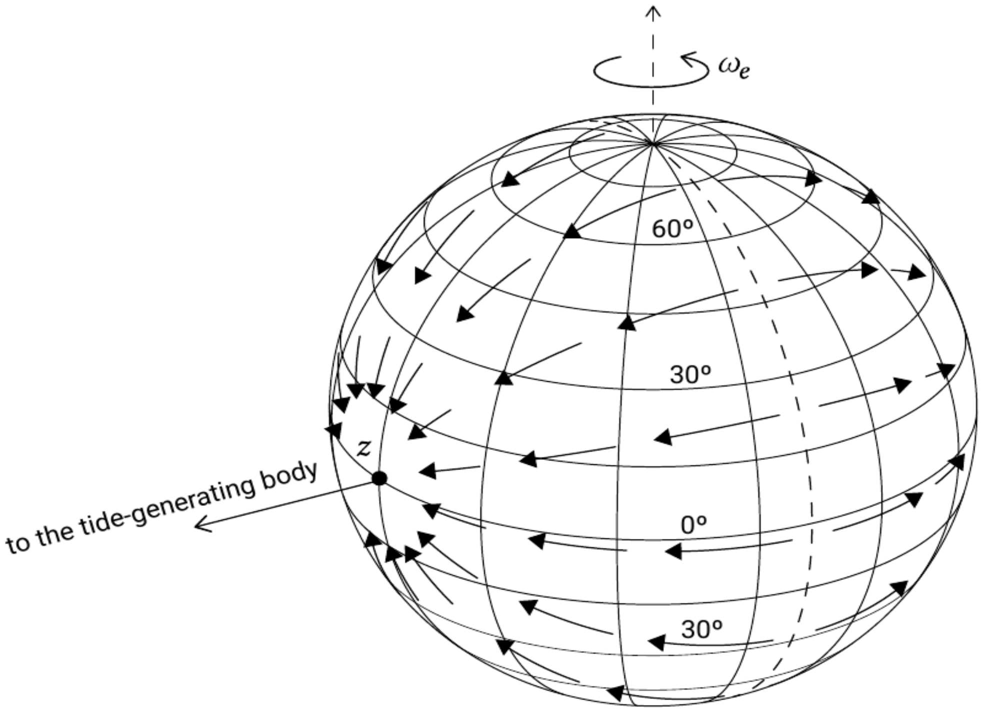


Figure 4: The horizontal component of the tidal force on the earth when the tide-generating body (the sun or the moon) is above the equator at z.

- **The effect of the tangential forces** is to **shift water to the side of the earth** facing the sun (moon) and to the opposite side in tidal bulges (see Figure 4).

- **The piling up of water** due to the tangential forces is **balanced by pressure gradients** in the opposite direction due to the sloping water surface.

- If the earth were completely covered by water, its equilibrium configuration would be an ellipsoid.

- **The rapid diurnal rotation** of the earth around its own axis makes the earth rotate underneath the tidal bulges, thus **producing a semi-diurnal tide** with two high and two low waters passing the same point on the earth every day.

- **The period of the solar tide is exactly 12** $\boldsymbol{h}$**.** The period of the lunar tide is governed by the period between the moon phases or the lunar day. **The period of the semi-diurnal lunar tide** is thus equal to half a lunar day = **12** $\boldsymbol{hours}$ **and 25** $\boldsymbol{minutes}$**.**

Problem

Using Newton's law of gravitation, compare the tide-generating force of the Moon with that of the Sun. The mass of the Sun is 2×10^{30} kg, the mass of the Moon is 7.35×10^{22} kg, the distance from Earth to Sun is 1.5×10^{8}km, and the distance from Earth to Moon is 3.84×10^{5} km.

Solution: Given $M_s = 2 \times 10^{30}$ kg, $M_m = 7.35 \times 10^{22}$ kg, $d_s = 1.5 \times 10^{8} km, d_m = 3.84 \times 10^{5} km$

As we know that tide-generating force is proportional to M/d^3, not M/d^2

$$F_{moon} / F_{sun} = (M_m/{d_m}^3) \,/\, (M_s/\,{d_s}^3)$$

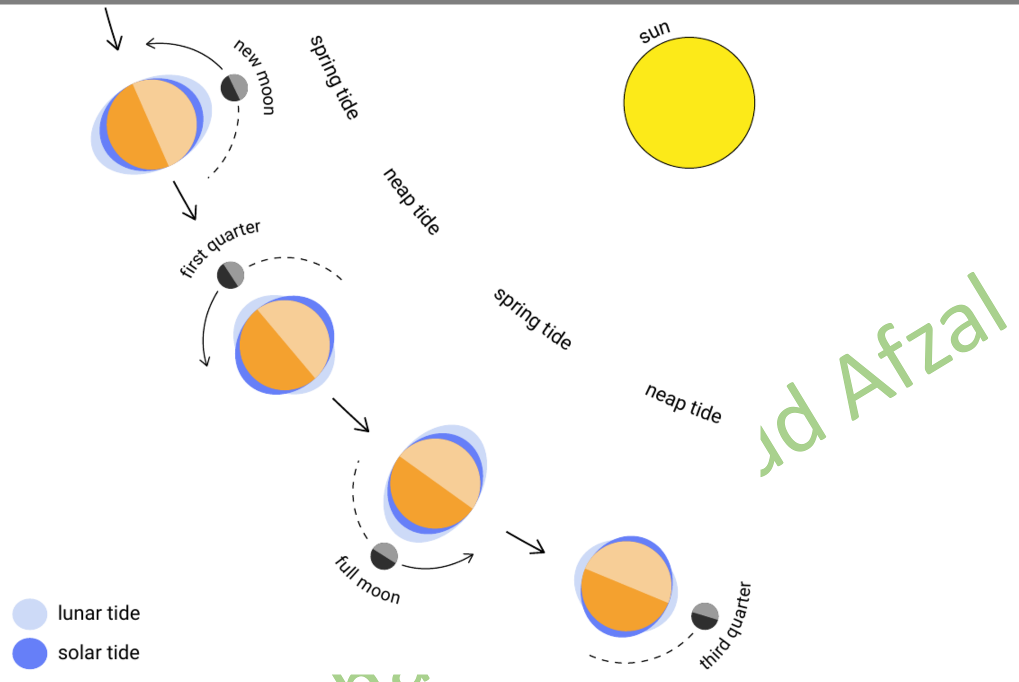
$$= (M_m/M_s) \times (d_s/d_m)^3$$

$$= (7.35 \times 10^{22}/2 \times 10^{30}) \times (1.5 \times 10^{8}/3.84 \times 10^{5})^3$$

$$= 2.19$$

The Moon's tide-generating force is approximately 2.19 times stronger than the Sun's. This explains why lunar tides dominate over solar tides.

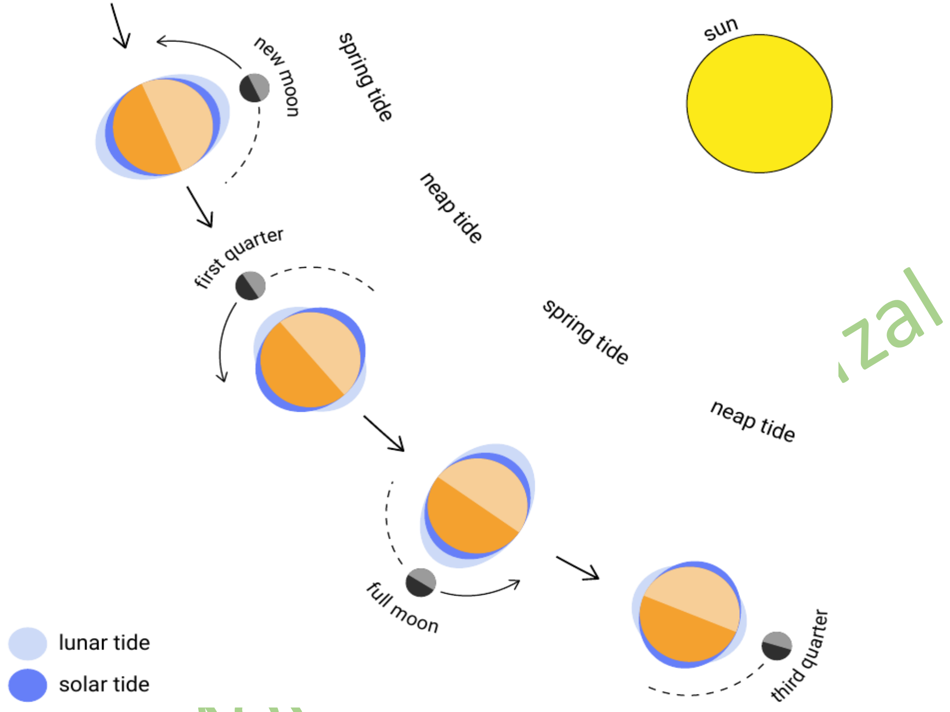
Spring tide



- When the **sun, the earth and the moon are in one line** (at full and new moon), the solar and lunar tides reinforce each other.

- The ellipsoid becomes more pronounced and the tide gets a **bigger amplitude**; this is called **spring tide.**

Neap tide



- When the solar and lunar tides are **90° out of phase**, their **effects cancel each other** (in the first and last quarter).

- The ellipsoid approaches a circle, and consequently the tide gets a **smaller amplitude**. This situation is called **neap tide**.

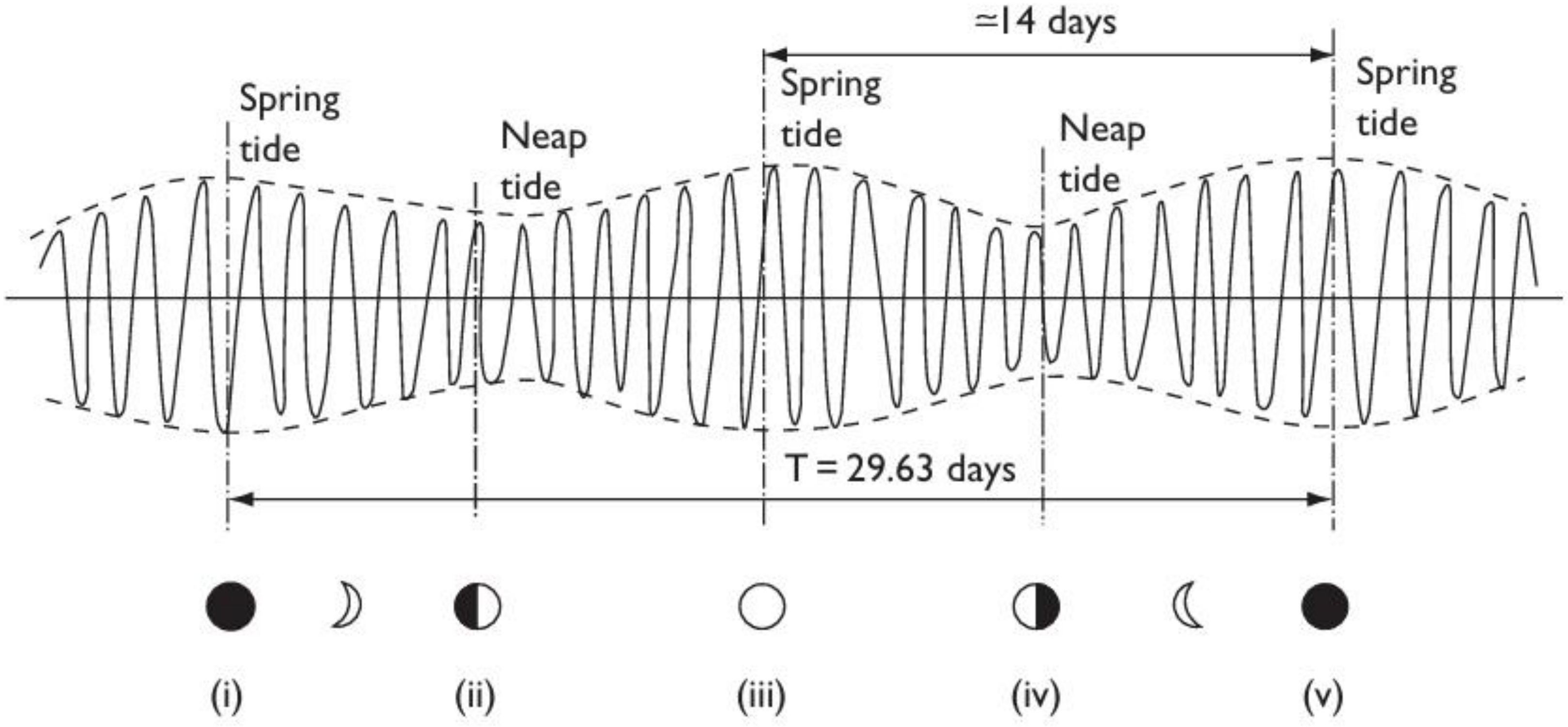


Figure 5: The Spring-neap variation in tidal range with phases of the Moon.

- The spring and neap tide cycle varies with moon phases and therefore with the lunar month of $29.63\ days$.

Tidal Components

- The **principal lunar** (M_2) **and solar** (S_2) **components** are just two of over 390 active tidal components having periods ranging from about 8 hours to 18.6 years.

- The principal **lunar tide** has a period of about **12.42 hours**, while the principal **solar tide** has a period of **12 hours**.

- **Each component** has a period that has been determined from **astronomical analysis** and **a phase angle and amplitude** that depend on local conditions and are best **determined empirically**.

- Most of the **390 components** are **quite small** and can be **neglected** for practical tide prediction.

- **Eight major components** with their common symbol, period, approximate relative strength (depending on location), and description are listed in **Table 1**.

Table 1: Eight Major Tidal Components

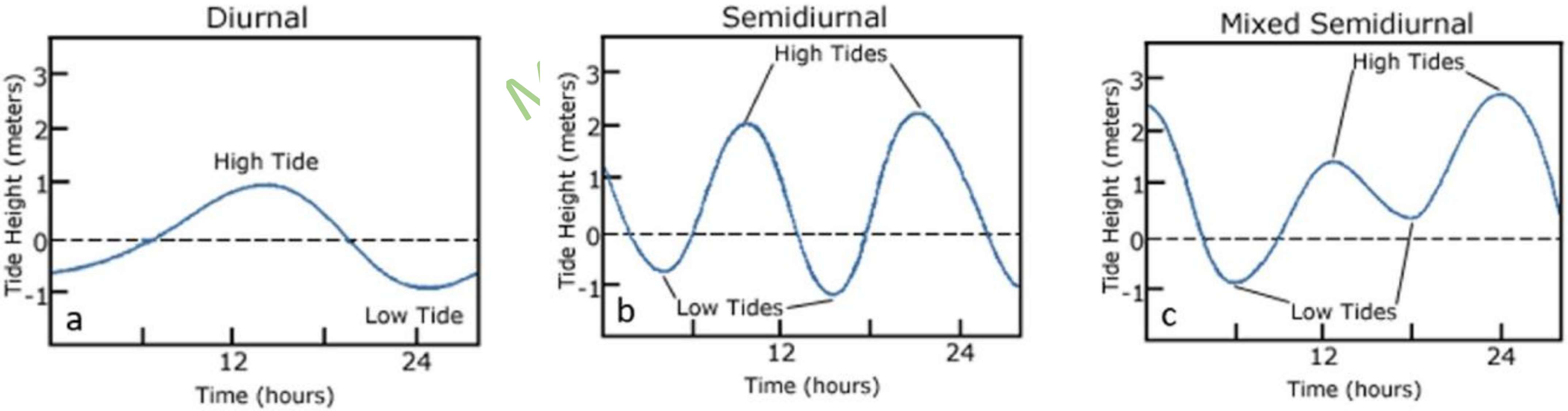
Symbol	Period (hours)	Relative Strength	Description
M_2	12.42	100.0	Main lunar semidiurnal component
S_2	12.00	46.6	Main solar semidiurnal component
N_2	12.66	19.1	Lunar component due to monthly variation in moons distance from earth
K_2	11.97	12.7	Soli−lunar component due to changes in declinatior of sun and moon throughout their orbital cycles
K_1	23.93	58.4	With O_1 and P_1 accounts for lunar and solar diurnal inequalities
O_1	25.82	41.5	Main lunar diurnal component
P_1	24.07	19.3	Main solar diurnal component
M_f	327.86	17.2	Lunar biweekly component

The tidal record at any location can generally be classified as one of three types:

- ***Diurnal:*** Occur once a day with **one high tide and one low tide**. Typically found in areas like Gulf of Mexico.

- ***Semidiurnal:*** Occur twice a day with **two high tides and two low tides** of approximately equal height. Commonly found along the Atlantic coast of North America and the Bay of Bengal.

- ***Mixed:*** Occur twice a day with **two high tides and one low tide** or **two low tides and one high tide.** Generally found along the western coast of North America.



Tidal ratio

- The **Tidal Ratio (F or N_f)** is a dimensionless number that expresses the relative importance of the **diurnal (once-daily)** tidal components compared to the **semidiurnal (twice-daily)** components at a specific location.

- The **Tidal Ratio**, also called the **Form Number** (F or N_f). It is expressed as :

$$N_f = \frac{K_1 + O_1}{M_2 + S_2}$$

$N_f < 0.25$	*Semidiurnal*	*Note: for mixed Tide*	
$0.25 < N_f < 3.0$	*Mixed*	$N_f < 1.5$	it is predominantly Semidiurnal
$N_f > 3.0$	*Diurnal*	$N_f > 1.5$	it is predominantly *Diurnal*

Problem

The tidal constituents for four harbours are given in the following table. Classify the tidal regime at each harbour using the tidal ratio.

Constituent	Period (h)	Harbour A Amplitude (cm)	Harbour B Amplitude (cm)	Harbour C Amplitude (cm)	Harbour D Amplitude (cm)
M_2	12.42	233	53	22	3
S_2	12.00	68	14	7	4
K_1	23.93	15	35	32	70
O_1	25.83	17	26	26	68
Z_0	-	0	50	0	-10

Solution: calculation of tidal ratio N_f from the amplitudes of the constituents:

For Harbour A: $$N_f = \frac{K_1 + O_1}{M_2 + S_2} = \frac{15+17}{68+233} = 0.11, \ \textit{tide is Semidiurnal}$$

For Harbour B: $N_f = \frac{K_1 + O_1}{M_2 + S_2} = \frac{26+35}{14+53} = 0.91,\ tide\ is\ Mixed, But\ predominantly\ Semidiurnal$

For Harbour C: $N_f = \frac{K_1 + O_1}{M_2 + S_2} = \frac{26+32}{22+7} = 2.0,\ tide\ is\ Mixed, But\ predominantly\ Diurnal$

For Harbour D: $N_f = \frac{K_1 + O_1}{M_2 + S_2} = \frac{68+70}{3+7} = 19.71,\ tide\ is\ Diurnal$

Daily inequality

- We have **either explicitly or implicitly** assumed that the sun and the moon are directly **above the equator of the earth.**

- Under that assumption a certain place on the earth experiences **two high waters and low waters per day of equal height** as can be seen from Figure 6.

- The heights of those high waters and low waters **depend on the latitude**.

- **In reality**, the orbits of the moon about the earth and of the earth about the sun are **not in the equatorial plane**, which complicates our simple picture of Figure 6.

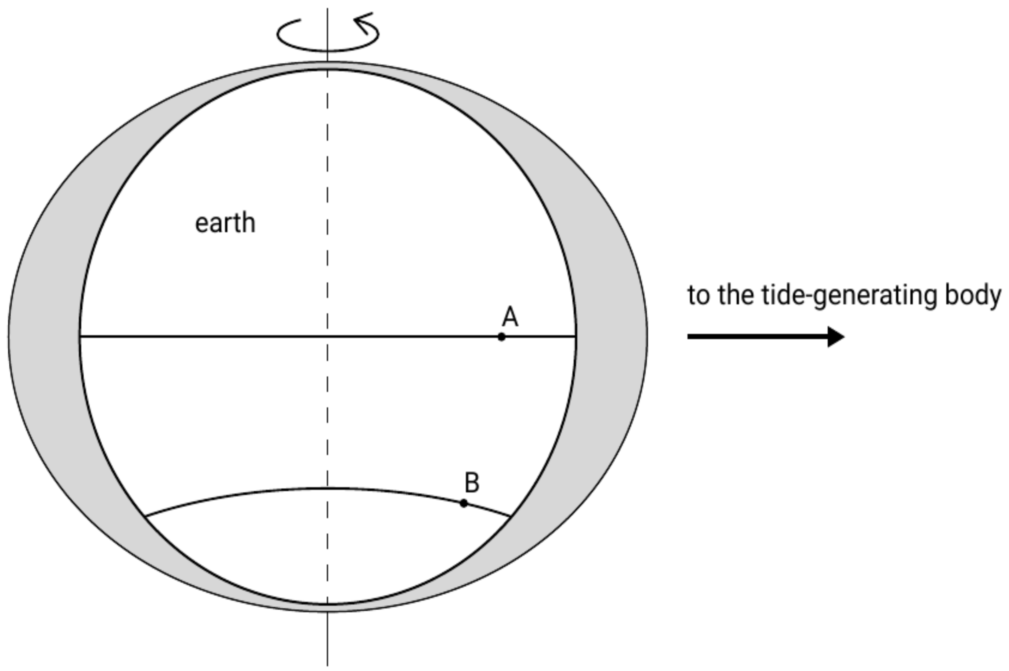


Figure 6: The tidal bulges for a tide-generating body above the equator. Observers *A* and *B* both experience two high and low waters a day of equal height. The heights depend on the latitude.

- While the moon's orbit and the earth's orbit are approximately in the same plane **(5° difference),** there is a **time-varying declination angle** between the equatorial plane and the earth-sun and earth-moon connection lines.

- As the **tidal bulges** tend to **align themselves** with the **tide-generating body**, the two high and low waters per day are **not equal** (see Figure 7). This phenomenon is referred to as **daily inequality.**

- Apparently, the declination of the sun has a diurnal (daily) effect on the tides. According to Figure 7, the daily **inequality is zero at the equator** and increases with latitude.

- At some **higher latitudes**, the **daily inequality becomes so big** that there is only one high and one low water.

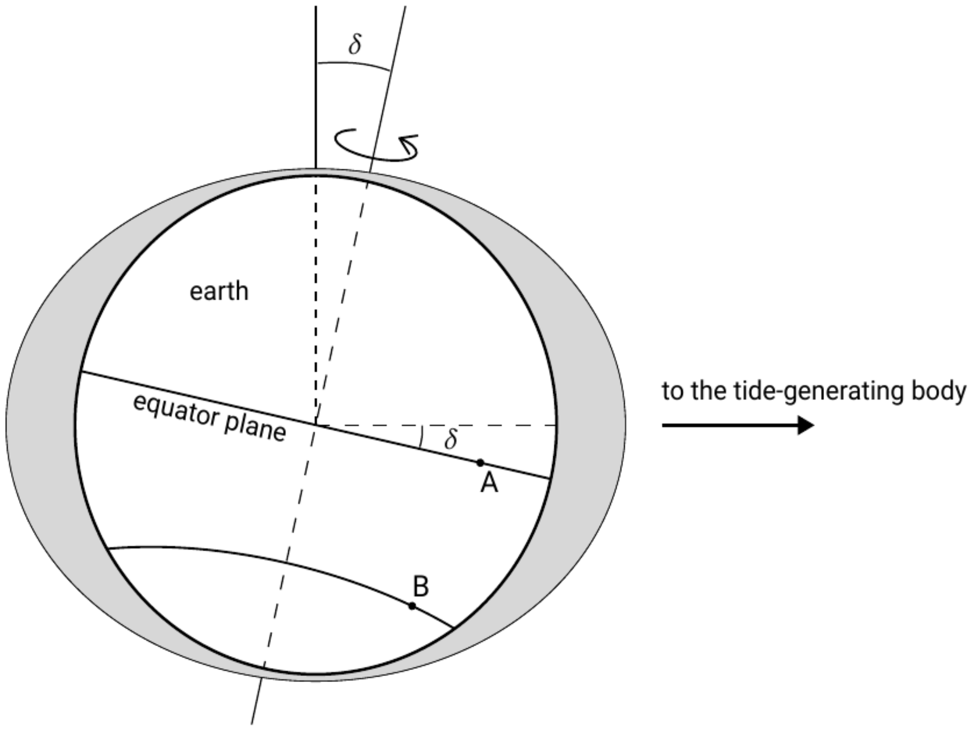


Figure 7: Daily inequality due to a non-zero declination angle δ between the equatorial plane and the line connecting the centers of the earth and the tide-generating body. For observer A the two high and low waters have the same height, but observer B experiences unequal heights in two successive high and low waters.

- The Earth's axis is **tilted by about 23.5°,** causing the Sun's declination **to vary seasonally.**
- It is **zero at the equinoxes** (March 22 and September 22) and positive in the **Northern summer**, and negative in the **Northern winter**, reaching **+23.5° on June 22** and **−23.5° on December 22**.
- This variation produces a **yearly cycle in solar tidal inequality**, with stronger spring tides near the equinoxes.
- Similarly, the **Moon's declination changes over a 27.3-day cycle**, producing lunar daily tidal inequality, which is greatest when the Moon reaches its maximum north or south position.

Propagation of the tide

Dynamic theory of tides:

- So far, the Earth has been assumed to be **completely covered by water**, with the planet rotating beneath a slowly varying tide.

- For a **semi-diurnal tide**, this is equivalent to a tidal wave travelling around the Earth once per day.

- In reality, **continents and limited ocean depth prevent the formation of a perfect equilibrium tide**, and land masses modify the movement of ocean water.

- This becomes clear when considering the **propagation properties** of the tidal wave.

- **The tidal wave is a long wave** because the wavelength $L \gg h$ and has a small amplitude (order $1\ m$ on an average ocean water depth of $4000\ m$).

- Therefore, and if friction can be neglected, the **wave propagation speed** follows from

$$c = \sqrt{gh} \qquad (5)$$

where, $c =$ wave propagation speed

$g =$ gravitation acceleration

$h =$ water depth

- For the **development of the equilibrium tide**, the semi-diurnal tide should cover **half of the circumference of the earth** in one period of $12\ hours$ and $25\ minutes$.

- The **circumference at the equator** is $2\pi R = 2\pi \times 6.37 \times 10^3\ km = 4.00 \times 10^4\ km$ and therefore the required propagation speed is $c = 4.00 \times 10^4\ km/(2 \times 12.42\ h) = 447\ m/s$.

- This means that the **tidal wave requires a water depth of 20 km much more** than the depth of the oceans – to travel fast enough.

- **The travel distance reduces towards the poles** and therefore the required propagation speed and water depth as well.

- The result is that the water depth becomes less of a limitation at higher latitudes.

Tidal propagation in the open oceans

- The tide can be described by the **shallow-water equations** (shallow-water approximation to the Navier-Stokes equations).

- For the tidal propagation in open oceans we can simplify these equations by **neglecting advection, friction, horizontal diffusion and short-wave effects.**

- We also assume a small tidal amplitude and bottom slope. **The resulting momentum equation** is a balance between acceleration (inertia) and the pressure gradient:

$$\frac{\partial u}{\partial t} = -g\frac{\partial \eta}{\partial x} \qquad (6)$$

where u = the horizontal velocity and

η = the oscillatory surface elevation.

- We further have the vertically integrated continuity equation:

$$\frac{\partial \eta}{\partial t} = -h\frac{\partial u}{\partial x} \qquad (7)$$

where h = the water depth.

- Eqs. 6 and 7 together describe the tidal propagation.
- Taking $\frac{\partial}{\partial t}$ of Eq. 7 and subtracting $h\frac{\partial}{\partial x}$ of Eq. 6 gives the classical wave equation:

$$\frac{\partial^2 \eta}{\partial t^2} = gh\frac{\partial^2 \eta}{\partial x^2} \qquad (8)$$

- Substitution of a progressive sine (or equivalently: cosine) wave $\eta = a\sin(\omega t - kx)$ into Eq. 8 yields:

$$\omega^2 = ghk^2 \qquad (9)$$

and thus (only taking the positive solution for c):

$$c = \frac{\omega}{k} = \sqrt{gh} \qquad (10)$$

- The velocity is in phase with the surface elevation:

$$u = \frac{gak}{\omega}\sin(\omega t - kx) \qquad (11)$$

- Evidently, for progressive waves the maximum velocities are found under the crest of the wave. The velocity amplitude is:

$$\hat{u} = \frac{gak}{\omega} = \frac{a\omega}{kh} = a\sqrt{\frac{g}{h}} \qquad (12)$$

- It is identical to the linear wave theory.

Amphidromic systems

- The **propagation of tides is influenced by friction, basin geometry, and ocean depth**.
- Because **tides occur on a very large scale**, their motion is also affected by the **Coriolis force**.
- As tidal waves are deflected by Coriolis and constrained by land masses, **rotary tidal motions develop in ocean basins**—counter-clockwise in the **Northern Hemisphere** and clockwise in the **Southern Hemisphere**.
- These rotating tidal systems are called **amphidromic systems**.

- In **an amphidromic system** the wave progresses about a node (no vertical displacement) with the antinodes (maximum vertical displacement) rotating about the basin’s edges (see Figure 8 and 9).

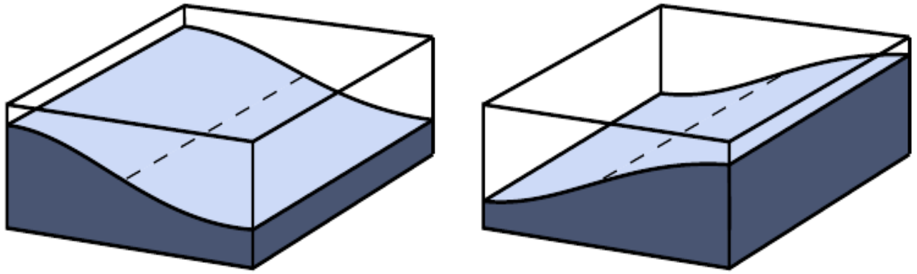


Figure 8: Normal standing wave.

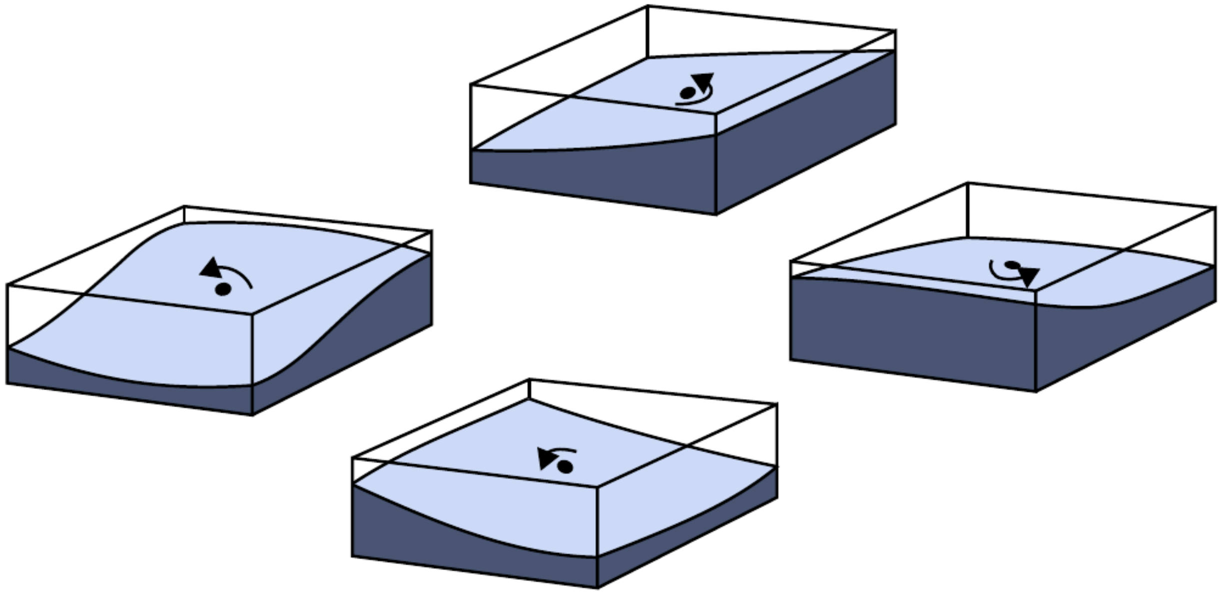
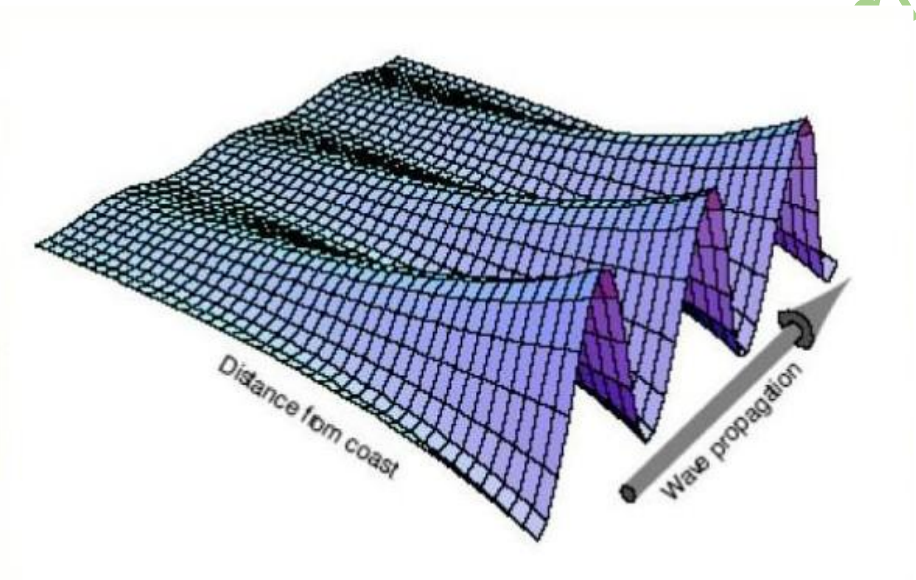


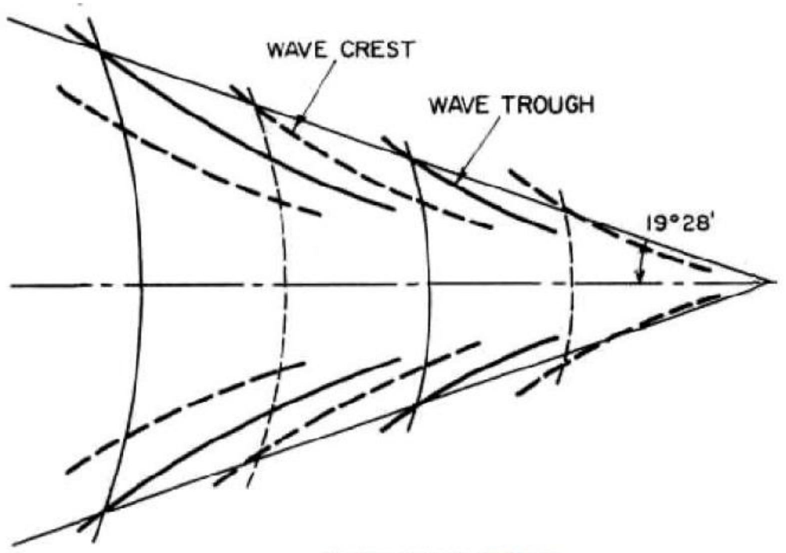
Figure 9: Tidal oscillation of a basin (Northern Hemisphere). The frequency of the oscillation is determined by the size of the basin and its depth.

- The water can be seen to be **sloshing around the basin**. The node, where the amplitude of the vertical tide is zero, is called an **amphidromic point**.

Kelvin waves

- A **Kelvin wave** is a **coastal trapped tidal wave** that propagates along a coastline due to the combined effects of **gravity and the Coriolis force**.
    - **Gravity:** Acts as the restoring force for the wave.
    - **The Coriolis Effect:** Causes the water to push against the coast as it travels, which keeps the wave trapped.
- These waves occur when tidal motion interacts with **coastal boundaries or ocean basin edges**.





Kelvin wave pattern

- Kelvin Wave is **derived** from the **shallow-water equations** by incorporating the **Coriolis effect** and the **presence of a vertical boundary** (the coastline).

- Assuming a straight coastline along the y-axis and a coordinate system where y is alongshore and x is cross-shore:

Alongshore Momentum (y): $$\frac{\partial v}{\partial t} = -g\frac{\partial \eta}{\partial y}$$

Cross-shore Momentum (x): $$-fv = -g\frac{\partial \eta}{\partial x}$$

Continuity Equation: $$\frac{\partial \eta}{\partial t} + h\left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}\right) = 0$$

Where, u, v velocity in $x, y-$direction

η oscillatory water level variation

f Coriolis parameter ($f = 2\omega \sin\varphi$)

h Mean water depth

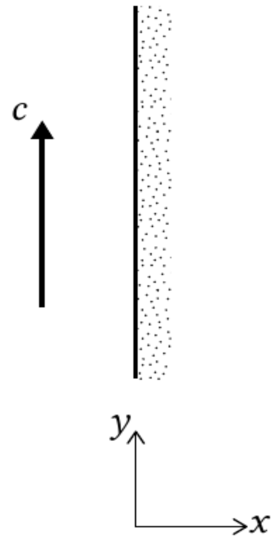


Figure 10: Closed eastern boundary with direction of tidal wave propagation (NH).

- By solving the above system for a wave propagating in the y-direction, we obtain the following expressions for surface elevation and velocity:

Surface Elevation (η):

$$\eta(x, y, t) = \eta_0 e^{\left(\frac{fx}{c}\right)} \cos(\omega t - ky)$$

- Surface elevation (η) describes a wave that moves in the y-direction while **decaying exponentially in the x-direction.**

Alongshore Velocity (u):

$$v(x, y, t) = \frac{c}{h} \eta_0 e^{\left(\frac{fx}{c}\right)} \cos(\omega t - ky)$$

- The water particles move **only parallel to the coast**, with a magnitude directly proportional to the wave height.

- **Rossby Radius of Deformation (R) or Offshore decay scale:** This defines the "trapping" distance—how far the wave extends offshore before its height drops significantly.

$$R = \frac{c}{f} = \frac{\sqrt{gh}}{f}$$

Main Features of Kelvin Waves :

- **Trapped Nature:** The wave amplitude (height) is at its maximum exactly at the coastline and decays exponentially as you move offshore.
- **No Cross-Shore Flow:** Because the Coriolis force is exactly balanced by the pressure gradient created by the water piling up against the coast (geostrophic balance), the water particles only move **alongshore**, never toward or away from the shore.
- **Phase Speed:** They travel at the same speed as a shallow-water gravity wave, calculated as

$$c = \sqrt{gh}.$$

Problem

A Kelvin wave propagates along a coastline with water depth of 80 m ($\omega = 72.9 \times 10^{-6}$rad/s). Calculate:

a) The phase speed,

b) Rossby Radius of Deformation (R) or Offshore decay scale at latitude 40°N

c) The time to travel 1000 km.

Solution: a) The phase speed $c = \sqrt{gh} = \sqrt{9.81 \times 80} = 28.01 m/s$

b) Rossby Radius of Deformation (R) or Offshore decay scale at latitude 40°N

$$R = \frac{c}{f}$$

$$f = 2\omega \sin\varphi = 2 \times 72.9 \times 10^{-6} \times \sin 40° = 9.33 \times 10^{-5} s^{-1}$$

$$R = \frac{28.01}{9.33 \times 10^{-5}} = 298.9km$$

c) The time to travel 1000 km

$$t = \frac{1000km}{28.01m/s \times 3.6} = 9.92hours$$